

Informative Power-System Probing via Sequential Bayesian Experimental Design

Gaoming Lin
*Electrical and Computer
Engineering
Texas A&M University
College Station, TX, USA
g.lin@tamu.edu*

Jinwoo Go
*Computational Science
Brookhaven National
Laboratory
Upton, NY, USA
jgo@bnl.gov*

Xiaoning Qian
*Electrical and Computer
Engineering
Texas A&M University
College Station, TX, USA
Computational Science
Brookhaven National
Laboratory
Upton, NY, USA
xqian@tamu.edu
xqian1@bnl.gov*

Byung-Jun Yoon
*Electrical and Computer
Engineering
Texas A&M University
College Station, TX, USA
Computational Science
Brookhaven National
Laboratory
Upton, NY, USA
bjyoon@tamu.edu
byoon@bnl.gov*

Abstract—Knowledge of latent grid parameters, including inertia and frequency-response coefficients, is important for assessing power-system stability. Active probing identifies these parameters from responses to injected signals. Fixed strategies predefine the probes without incorporating new measurements, while myopic strategies optimize one probe at a time and can require expensive online computation. We formulate active probing as a sequential Bayesian experimental design problem and use learned adaptive policies to maximize information about the latent parameters. In experiments on reduced IEEE 9-bus and IEEE 14-bus models, deep adaptive design (DAD), Step-DAD, and reinforcement learning (RL-sBOED) achieve higher mean sPCE lower-bound information scores than myopic, optimized fixed-sequence, and random probing under the same probe budget. On the IEEE 9-bus model, DAD exceeds Fixed by 0.09–0.13 nats in mean sPCE score. DAD and RL-sBOED select probes with trained network evaluations, whereas Myopic and Step-DAD require additional online simulation and optimization. These results support further study of learned sequential policies for informative grid probing.

Index Terms—Bayesian experimental design, power-system identification, adaptive probing, expected information gain, rate of change of frequency.

I. INTRODUCTION

As inverter-based resources displace synchronous generation, changing inertia and frequency response increase the need to characterize grid dynamics [1]. Measurement-based inertia estimation uses frequency and active-power responses to disturbances [2]; controlled probing offers an alternative when naturally occurring events do not provide suitable excitation. Field tests on the western North American interconnection demonstrate that probing-signal design affects the quality of electromechanical-mode estimates [3], and recent work applies probing to inertia and frequency-response estimation in systems with inverter-based resources [4].

The design of informative excitation is a central problem in system identification [5]. Here, we study a narrower question: how should successive pulse durations adapt to measurements

when each probe begins from the physical state left by the previous probe? Duration changes both the injected energy and the ensuing transient. Because the state is not reset, a decision also changes the conditions under which later measurements are collected. Selecting a sequence therefore requires accounting for both immediate information and future experiments.

Sequential Bayesian optimal experimental design provides a framework for this tradeoff [6]. A fixed sequence cannot respond to incoming measurements, whereas a myopic design optimizes only the next observation. Deep adaptive design (DAD) [7] and reinforcement-learning-based sequential design (RL-sBOED) [8] learn history-dependent policies using a full-sequence information objective. Step-DAD [9] additionally refines a pretrained policy during the experiment, trading online computation for further adaptation.

We formulate duration selection for reduced IEEE 9-bus and IEEE 14-bus swing models with unknown machine inertia and effective frequency-response coefficients. Our contributions are a carried-state Bayesian probing formulation, a comparison of six methods under a common restricted duration space and probe count, and an analysis of information scores, selected durations, and algorithmic computational requirements. The study evaluates the transfer of sequential-design methods to this grid benchmark; it does not establish unrestricted optimal probing, operational safety, or complete identification from a single sensor.

II. PROBLEM FORMULATION

We seek to reduce uncertainty about grid parameters using active probing within a budget of T design steps. Here $t = 1, \dots, T$ indexes design steps, d denotes elapsed time within each probe window, and D_t is the selected pulse duration.

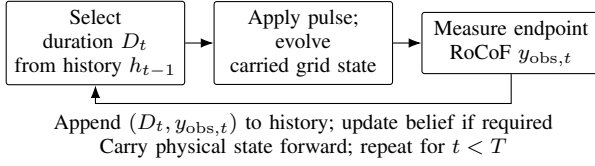


Fig. 1. Sequential probing loop. Adaptive methods use observation history; Fixed and Random select durations without observation feedback.

A. Latent Parameters and Power-System Model

For a reduced network with m machines, the latent parameter vector is $\theta = (M_1, \dots, M_m, K_1, \dots, K_m)$, where M_i and K_i are inertia and effective frequency-response coefficients. The prior $p_0(\theta)$ has independent components $M_i \sim \mathcal{U}(0.7\bar{M}_i, 1.3\bar{M}_i)$ and $K_i \sim \mathcal{U}(0.1\bar{K}_i, \bar{K}_i)$. One draw of θ remains fixed throughout all T probes.

Let δ_i be rotor angle, ω_i angular-speed deviation, and u_i the applied probing power. The reduced swing model is

$$\begin{aligned} \dot{\delta}_i &= \omega_i, \\ M_i \dot{\omega}_i &= u_i - \left(D_i + \frac{K_i}{2\pi} \right) \omega_i \\ &\quad - \sum_{j=1}^m B_{ij} \sin(\delta_i - \delta_j). \end{aligned} \quad (1)$$

Dots denote derivatives with respect to physical time d . In (1), M_i scales acceleration, while K_i contributes a restoring power response proportional to frequency deviation. It is an effective coefficient, not an explicit turbine-governor model. The damping D_i and network couplings B_{ij} are known; frequency deviation is $\Delta f_i = \omega_i/(2\pi)$ in Hz. The units of M_i , K_i and D_i are pu s²/rad, pu/Hz and pu s/rad, respectively, on a 100-MVA base at nominal frequency 60 Hz. All angles and speed deviations start at zero. Net mechanical injections are zero in this deviation model, making the initial unforced state an equilibrium.

The reduced IEEE 9-bus model retains buses 1, 2, and 3; the reduced IEEE 14-bus model retains buses 1, 2, 3, 6, and 8. Test-case reactances [10] define a lossless Laplacian $\mathbf{L}$. Eliminating algebraic buses l while retaining machines g gives $\mathbf{L}_{\text{red}} = \mathbf{L}_{gg} - \mathbf{L}_{gl} \mathbf{L}_{ll}^{-1} \mathbf{L}_{lg}$, with $B_{ij} = -(\mathbf{L}_{\text{red}})_{ij}$ for $i \neq j$ and $B_{ii} = 0$. Resistance, voltage dynamics, and transformer tap effects are omitted. Table I summarizes the model parameterization. Nominal inertia is computed as $\bar{M}_i = 2H_i/(2\pi f_0)$, with $f_0 = 60$ Hz. For IEEE 9-bus, $H = (23.64, 6.40, 3.01)$ s matches published machine data [11, Table 2]. For IEEE 14-bus, $H = (4, 6.5, 5, 5, 5)$ s is prescribed at buses (1, 2, 3, 6, 8) as assumed benchmark values. For both networks, $\bar{K}_i = K_{\text{tot}} w_i$, with $K_{\text{tot}} = 0.5$ pu/Hz. The allocation weights are $w = (0.5875, 0.2625, 0.15)$ for IEEE 9-bus and $w = (8, 13, 10, 10, 10)/51$ for IEEE 14-bus. Fixed damping satisfies $D_i = (0.1 \text{ s}^{-1}) \bar{M}_i$. The response scales, damping values, and prior ranges are benchmark assumptions rather than validated operational grid parameters.

TABLE I
PRIOR SCALES AND DAMPING COEFFICIENTS

Network	Bus	$\bar{M}_i$	$\bar{K}_i$	D_i
IEEE 9-bus	1	0.1254141	0.2937500	0.0125414
	2	0.0339531	0.1312500	0.0033953
	3	0.0159685	0.0750000	0.0015969
IEEE 14-bus	1	0.0212207	0.0784314	0.0021221
	2	0.0344836	0.1274510	0.0034484
	3, 6, 8	0.0265257	0.0980392	0.0026526

B. Probe Design and Observation Model

The probe design is $\xi_t = (A_t, B_t, D_t)$: amplitude, bus location, and probe signal duration. This study fixes $A_t = 0.05$ pu and $B_t = 1$; only D_t is optimized. The raised-cosine input for $0 \leq d \leq D_t$ is

$$u_{B_t,t}(d) = \frac{A_t}{2} [1 - \cos(2\pi d/D_t)]. \quad (2)$$

The input is zero for $D_t < d \leq D_{\text{obs}}$ and at all other buses. Fixing amplitude and location isolates duration selection rather than joint actuator design. The pulse injects energy proportional to $A_t D_t/2$; equal probe counts therefore do not imply equal injected energy. The observation window is $D_{\text{obs}} = 3.5$ s. The full angle and speed state at its end becomes the initial state of the next window, without resetting the system. Consequently, the response depends on the complete design sequence $\xi_{1:t} = (\xi_1, \dots, \xi_t)$.

Write $\Delta f_{1,t}(d; \theta, \xi_{1:t})$ for the resulting bus-1 frequency deviation. The noise-free response is the signed endpoint rate of change of frequency (RoCoF), denoted by y_t :

$$y_t(\theta, \xi_{1:t}) = \frac{\Delta f_{1,t}(D_{\text{obs}}) - \Delta f_{1,t}(D_{\text{obs}} - \Delta d_{\text{samp}})}{\Delta d_{\text{samp}}}. \quad (3)$$

The frequency arguments $\theta, \xi_{1:t}$ are suppressed in the numerator. With sampling interval $\Delta d_{\text{samp}} = 0.025$ s, the measured response follows

$$y_{\text{obs},t} = y_t(\theta, \xi_{1:t}) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (4)$$

where $\sigma = 0.005$ Hz/s and noise is independent across windows. The endpoint measurement summarizes the post-pulse transient, rather than the initial inertial response. A single fixed sensor keeps the observation model simple and identical across methods, but discards information in the full frequency trajectory and at other buses. This is a benchmark choice, not an optimized sensor configuration. Each observation becomes available at window end. Decision latency is idealized as zero in simulated time.

C. Sequential Decisions and Bayesian Updating

The data after step t are the ordered design-observation history $h_t = ((\xi_k, y_{\text{obs},k}))_{k=1}^t$, with $h_0 = \emptyset$. A policy π selects ξ_t using only observed history h_{t-1} , with amplitude and bus fixed as above. It has no access to true parameters or physical state. Feasible durations satisfy

$$\begin{aligned} 0.2 &\leq D_t \leq 3, \quad t = 1, \dots, T, \\ D_t - D_{t-1} &\geq \Delta D_{\text{min}}, \quad t = 2, \dots, T, \end{aligned} \quad (5)$$

The increasing-duration restriction defines the benchmark design space; it is not a physical requirement of the swing equations. It was adopted to retain the ordered-design structure of an earlier sequential-design setup. Applying it to every method permits comparisons within this space, but excludes nonmonotonic sequences that may be more informative. Thus, optimality in (8) is restricted to Π . Here, $\Delta D_{\min} = 0.01$ s. At step t , the upper bound is $3 - (T - t)\Delta D_{\min}$ to leave enough room for the remaining probes. The lower bound starts at 0.2 s and then becomes $D_{t-1} + \Delta D_{\min}$. All methods map a unit action to this interval without sorting or discretizing durations. Let Π denote the set of policies that satisfy these constraints; the policy randomness remains independent of θ .

The Gaussian likelihood $p(y_{\text{obs},t} \mid \theta, h_{t-1}, \xi_t)$ has mean $y_t(\theta, \xi_{1:t})$ and variance σ^2 . Bayes' rule updates parameter uncertainty after each measurement:

$$p(\theta \mid h_t) = \frac{p(y_{\text{obs},t} \mid \theta, h_{t-1}, \xi_t) p(\theta \mid h_{t-1})}{p(y_{\text{obs},t} \mid h_{t-1}, \xi_t)}. \quad (6)$$

The denominator is the predictive observation density, obtained by integrating the numerator over θ , with $p(\theta \mid h_0) = p_0(\theta)$. Each candidate parameter vector propagates its corresponding physical state under the observed designs to evaluate the likelihood. Conditional on the observed history, design selection provides no additional information about θ , so policy probabilities cancel from the Bayesian update.

D. Expected Information Gain and Design Objective

To compare possible design policies, we quantify the uncertainty reduction achieved by their observations using the Kullback–Leibler divergence between the terminal posterior and the prior. The expected information gain (EIG) of policy π is

$$\mathcal{I}_T(\pi) = \mathbb{E}_{h_T \mid \pi} [D_{\text{KL}}(p(\theta \mid h_T) \parallel p_0(\theta))]. \quad (7)$$

Here $D_{\text{KL}}(p \parallel q) = \int p(\theta) \log[p(\theta)/q(\theta)] d\theta$; information gain is measured in nats. The expectation averages over parameter draws, measurement noise, and any policy randomness. The sequential Bayesian design problem is

$$\pi^* \in \arg \max_{\pi \in \Pi} \mathcal{I}_T(\pi). \quad (8)$$

Equation (8) defines the common information objective; the methods below differ in how they approximate or forgo its optimization. All receive the same number of probes. Computation is assessed separately.

Evaluating EIG requires expensive marginal-likelihood calculations. We therefore use the sequential prior-contrastive (sPCE) lower bound [7] for training and evaluation, instead of the true EIG. Let $\theta^{(0)} \sim p_0$ generate the observed history and let $\theta^{(1)}, \dots, \theta^{(L)}$ be independent contrastive samples from the prior. For any candidate θ , define the observation likelihood along the realized sequence as

$$\mathcal{L}_t(\theta) = \prod_{k=1}^t p(y_{\text{obs},k} \mid \theta, h_{k-1}, \xi_k). \quad (9)$$

The sampled information score is then

$$\hat{I}_t = \log \frac{\mathcal{L}_t(\theta^{(0)})}{\frac{1}{L+1} \sum_{\ell=0}^L \mathcal{L}_t(\theta^{(\ell)})}. \quad (10)$$

Likelihoods are evaluated in the log domain. Averaging $\hat{I}_T$ estimates the sPCE lower bound on $\mathcal{I}_T(\pi)$, not the exact EIG. Individual scores may be negative but are bounded above by $\log(L+1)$. With $L = 128$, the ceiling is 4.8598 nats; near-ceiling scores can conceal differences in EIG. We report these estimated lower-bound values as information scores throughout the results. The generating parameters are used only to score simulated observations.

III. COMPARED DESIGN METHODS

All six methods use the same observation model and feasible duration interval (5). an adaptive method need not be myopic, and a fixed sequence can be optimized for the full horizon. The algorithms are adaptations of existing methods to the carried-state grid simulator.

A. Nonadaptive Designs: Random and Fixed

Random samples $z_t \sim \mathcal{U}(0, 1)$ independently at each step and sets $D_t = D_t^{\min} + (D_t^{\max} - D_t^{\min})z_t$. Here $D_1^{\min} = 0.2$, $D_t^{\min} = D_{t-1} + \Delta D_{\min}$ for $t > 1$, and $D_t^{\max} = 3 - (T - t)\Delta D_{\min}$. Thus, random samples are uniform within the remaining feasible interval and are not uniform over feasible complete sequences. Random does not use observations and needs no offline training.

Fixed is a static Bayesian design: all durations are optimized jointly before observations are collected [6, Sec. 2.4]. The ordered sequence is optimized using the full-sequence sPCE objective. It compares short, central, long, and spread interior initializations using validation, then optimizes the selected initialization. Each training seed yields a separately fitted sequence. During evaluation, Fixed executes that sequence without adapting to observations.

B. Online One-Step Design: Myopic

Unlike Fixed, Myopic uses observation feedback. It follows one-step adaptive BOED, maximizing the expected information from the next observation under the current posterior [6, Sec. 2.4]. Our implementation maintains a weighted particle approximation to this posterior. Before each decision after the first, particles propagate their own states under the preceding duration and update likelihood weights using the latest observation. Continuous search then maximizes expected information from the next probe only, using 16 proposals, three search rounds, and 32 simulated observations per proposal. Candidate durations obey the same constraints as all other methods. No offline policy is trained.

C. Offline-Trained Adaptive Methods: DAD and RL-sBOED

DAD and RL-sBOED use neural policies trained on simulated sequences drawn from the same parameter prior as Fixed. Both methods receive the stage index and ordered trajectories, including past durations, scaled observations, and availability

masks; future slots are masked. During evaluation, actions are generated via a forward pass without online optimization or explicit particle-posterior updates.

DAD [7] employs a deterministic two-layer, 64-unit tanh network, a trainable stage bias, and a sigmoid action map. Offline training maximizes sPCE by differentiating through the entire simulated sequence, including both observations and carried states. In joint Fixed/DAD training, validation determines whether to use random or interior Fixed-based initialization. Section V-C addresses the associated initialization cost.

RL-sBOED [8] uses a REDQ-style ensemble actor-critic framework with ten critics and five replay updates per trajectory batch. The actor consists of two 128-unit ReLU layers. Per-step information rewards are defined as $R_t = \hat{I}_t - \hat{I}_{t-1}$, with $\hat{I}_0 = 0$ and a discount factor $\gamma = 1$. As a result, the cumulative reward equals the terminal sPCE score. Offline training accounts for future design steps; online evaluation uses the trained actor without additional planning simulations. Entropy is regularized by normalized tanh actions and a target entropy of -1 ; reported information scores do not include entropy bonuses. We use the actor’s deterministic output for evaluation.

D. Offline-Trained Online-Refined Adaptive Methods: Step-DAD

Step-DAD [9] starts each episode from its selected DAD checkpoint and leaves the first probe unchanged. Before subsequent decisions, a 128-particle posterior supplies candidate parameters and their carried states for simulations of the remaining sequence. A fresh Adam optimizer performs four updates per decision (batch size 32, learning rate 10^{-3}). A fixed conditional validation batch selects the best candidate, including the unchanged policy. Retained weights persist within an episode only. The total refinement budget is $4(T - 1)$ updates. Myopic and Step-DAD require posterior assimilation only before subsequent decisions; terminal information scoring is separate.

IV. EXPERIMENTAL SETTINGS

A. Simulation and Training Protocol

For each network, we use three, four, or five probes, with one observation per probe. Both networks use the same sensor specification and design constraints but have different physical parameters and separately trained policies.

The fourth-order Runge-Kutta (RK4) method is used to integrate the swing equations, with a step size of $1/640$ s. The numerical state and duration Jacobians are calculated using perturbations of 10^{-5} and take into account the derivatives of the carried states. The validation suite compares the policy gradients and the conditional refinement gradients with those obtained by end-to-end finite differences.

Offline training uses 2000 updates, a batch size of 32, and three independent training seeds (101, 202, and 303). The DAD and Fixed learning rates are both 10^{-3} . Validation is carried out using 128 systems every 100 updates. For the sPCE

objective, 128 contrastive samples are used in the training, validation, and evaluation phases. The Myopic and Step-DAD methods each use 128 particles for online posterior planning, and Step-DAD carries out four refinement updates after each observation so long as design steps remain. The evaluation comprises seeds 1001, 1002, and 1003, with 512 simulated experimental sequences per evaluation seed. Validation cases are not used for gradient updates. Each trained policy is assessed over 1536 episodes. The draws for parameters and measurement noise are paired across the different methods, whereas the online planner makes use of its own separate random stream. Since the Random and Myopic methods show no dependence on the training seed under these streams, identical evaluations can be obtained for all training seeds. Fixed is optimized independently for each training seed.

B. Evaluation Protocol

For trained methods, we average the evaluation-seed means within each training seed and report the mean $\pm$ sample standard deviation across the three training seeds. For Random and Myopic, we report the mean $\pm$ sample standard deviation across evaluation-seed means rather than episode-level dispersion. Evaluation seed 1001 was used during development, so the evaluation is not fully independent.

Reported timing measurements use NVIDIA A100 PCIe 40-GB GPUs. IEEE 9-bus offline training used Xeon Platinum 8352Y CPUs; its new online timing measurements and the IEEE 14-bus measurements use Xeon Gold 6248R CPUs. Only the A100 timing subset is reported; timing coverage is specified in Section V-C. Code and saved results are available at https://github.com/alexschmidt123/tpec2027_power_grid.

V. RESULTS AND DISCUSSION

A. Information Acquired at Equal Probe Budgets

Tables II and III report mean sPCE information scores in nats, with the seed-level standard deviations defined above. All three learned adaptive methods have higher mean sPCE scores than Random, Fixed, and Myopic in the reported network-horizon comparisons.

TABLE II
IEEE 9-BUS INFORMATION SCORE (SPCE, NATS)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.4321 ± 0.0310	0.4924 ± 0.0342	0.5392 ± 0.0471
Fixed	2.0112 ± 0.0116	2.3192 ± 0.0004	2.5066 ± 0.0013
Myopic	1.9736 ± 0.0438	2.1707 ± 0.0366	2.2407 ± 0.0547
DAD	2.1366 ± 0.0161	2.4095 ± 0.0080	2.6351 ± 0.0179
RL-sBOED	2.1257 ± 0.0027	2.3963 ± 0.0333	2.5307 ± 0.0114
Step-DAD	2.1471 ± 0.0110	2.4382 ± 0.0048	2.6621 ± 0.0145

On IEEE 9-bus, DAD exceeds Fixed by 0.1253, 0.0903, and 0.1286 nats at $T = 3, 4, 5$, respectively. Step-DAD improves the DAD mean by 0.0105, 0.0287, and 0.0269 nats. These mean improvements are small; paired uncertainty estimates are needed to assess their reliability. The IEEE 9-bus information scores are well below the estimator ceiling.

TABLE III
IEEE 14-BUS INFORMATION SCORE (sPCE, NATS)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.7541 ± 0.0047	0.8360 ± 0.0297	0.8464 ± 0.0252
Fixed	4.2623 ± 0.0402	4.4946 ± 0.0345	4.6624 ± 0.0083
Myopic	4.2907 ± 0.0368	4.3990 ± 0.0177	4.4641 ± 0.0280
DAD	4.7576 ± 0.0362	4.8325 ± 0.0047	4.8443 ± 0.0026
RL-sBOED	4.7445 ± 0.0022	4.8275 ± 0.0112	4.8383 ± 0.0108
Step-DAD	4.7527 ± 0.0275	4.8297 ± 0.0067	4.8446 ± 0.0042

On IEEE 14-bus, learned-policy scores approach the 4.8598-nat ceiling, preventing reliable EIG rankings. Step-DAD changes the DAD mean by -0.0049 , -0.0028 , and $+0.0003$ nats at $T = 3, 4, 5$. Fixed exceeds Myopic at $T = 4, 5$ under the stated search budget; a larger Myopic budget could change this comparison.

B. Selected Probe Durations

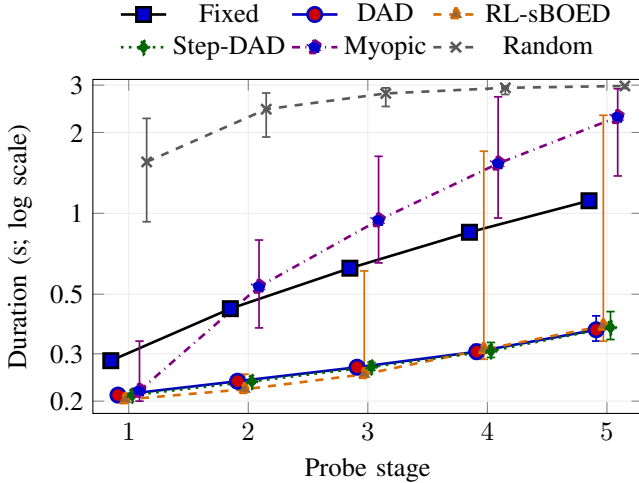


Fig. 2. IEEE 14-bus pulse durations at $T = 5$: medians and interquartile ranges for all six methods. Fixed and learned methods: 4608 histories over three seeds; Myopic and Random: 1536 each. Horizontal offsets separate methods. Bars show pooled variation, not confidence intervals; ordering is imposed.

At $T = 5$, median final durations are 0.369 s for DAD, 0.376 s for Step-DAD, 1.114 s for Fixed, 2.283 s for Myopic, and 2.979 s for Random (Fig. 2). Random’s sequential interval sampling concentrates later pulses near the upper bound. These patterns reflect different energy allocations, but do not isolate feedback effects from seed variation. Ordering is imposed; unrestricted and energy-matched comparisons remain open.

C. Computational Cost

Tables IV and V report available A100 PCIe 40-GB timings at $T = 3, 4, 5$, using training seed 101. Offline times describe one preparation workflow. IEEE 9-bus online times average 32 paired simulated episodes per method after one discarded warm-up episode; IEEE 14-bus online times use 1536 archived

episodes. Online timing includes posterior updates and refinement but excludes true-system simulation, observation acquisition, and information scoring. Offline runs use different CPU models, and measurements span cluster nodes; these are descriptive hardware-specific measurements. Let C_s be one candidate-window simulation cost and C_π a policy forward pass. Dense m -machine RK4 integration over N time steps gives $C_s = O(Nm^2)$. The theoretical expressions below identify the source of computational work.

Online policy work is $O(T)$ for Random/Fixed and $O(TC_\pi)$ for DAD/RL-sBOED. Myopic uses $O(TQC_e + TPC_s)$, where $P = 128$ particles and Q candidate objectives of cost C_e . Step-DAD adds remaining-horizon refinement proportional to $UBT(T-1)/2$, with $U = 4$ updates and $B = 32$ trajectories per update; numerical Jacobians and validation add work.

TABLE IV
IEEE 9-BUS RECORDED A100 TIMINGS

Method	$T = 3$	$T = 4$	$T = 5$
Offline preparation (hours)			
Random	0	0	0
Fixed	0.554	0.802	1.063
DAD	1.116	1.608	2.139
RL-sBOED	0.100	0.117	0.129
Myopic	0	0	0
Step-DAD	1.116	1.608	2.139
Online decisions (ms per sequence)			
Random	0.172	0.263	0.325
Fixed	1.032	1.489	1.550
DAD	1.182	1.716	2.062
RL-sBOED	1.178	1.823	2.241
Myopic	4414	5955	7560
Step-DAD	3233	7952	14553

Seed 101; DAD/Step-DAD include Fixed initialization.
Online: 32 episodes per method after one discarded warm-up.

Let C_g include per-window simulation and gradient work. Offline training with J updates, batch size B , and L contrasts requires $O(JB(L+1)TC_g)$ simulation/gradient work for Fixed and DAD. DAD’s complete preparation includes the Fixed stage: $W_F + W_D$; Step-DAD inherits both. Random and Myopic need no offline training. RL-sBOED uses $O(JB(L+1)TC_s)$ simulated trajectory work plus actor-critic replay updates and validation. Offline training times are single-run observations, not seed-averaged estimates.

Rescoring n histories costs $O(n(L+1)TC_s)$ simulation work plus $O(nLT)$ likelihood arithmetic, without retraining. These hardware-specific timings do not establish real-time feasibility; offline training uses different CPU models and online evaluation uses different sample sizes.

D. Estimator Sensitivity

The ceiling $\log(L+1)$ limits sPCE, not EIG. A mean below it can conceal saturated histories. Larger- L evaluation of the same saved histories is needed to assess this bias without retraining.

TABLE V
IEEE 14-BUS RECORDED A100 TIMINGS

Method	$T = 3$	$T = 4$	$T = 5$
Offline preparation (hours)			
Random	0	0	0
Fixed	1.880	2.750	3.720
DAD	3.810	5.457	7.418
RL-sBOED	0.163	0.205	0.249
Myopic	0	0	0
Step-DAD	3.810	5.457	7.418
Online decisions (ms per sequence)			
Random	0.161	0.209	0.309
Fixed	1.377	1.811	4.885
DAD	1.279	1.534	1.895
RL-sBOED	1.963	5.579	2.290
Myopic	10000	13437	16935
Step-DAD	9234	23357	45101

Seed 101; DAD/Step-DAD include Fixed initialization.

Reserved for IEEE 14-bus
larger- L evaluation results
 $T = 3, 4, 5$; $L = 10^4, 10^5$

E. Scope and Limitations

These conclusions apply to the specified reduced lossless dynamics, independent parameter priors, and scalar sensor. IEEE 14-bus has ten unknown parameters but only three to five scalar observations. The prior and nonlinear likelihood can reduce joint uncertainty, but the information score does not establish accurate identification of each M_i or K_i . Parameter-wise posterior error, interval width, and coverage are needed for that assessment. Monotonic durations, fixed probe location, and the endpoint sensor further restrict the conclusions. The same parameter distribution is used during both training and evaluation; we have not demonstrated robustness with respect to prior shift, sensor bias, or unmodelled dynamics. Evaluation seed 1001 was used during development, so the results do not provide fully independent confirmation. The errors introduced by finite-contrast scoring and particle inference are different from those caused by ODE integration. Lastly, the zero-latency simulation does not prove that online planners are feasible in real time, and increased information gain on its own does not show that control is safer.

VI. CONCLUSION

Learned adaptive policies use observation history to select probes over the experimental sequence. When three to five probes are used on the reduced IEEE 9-bus and IEEE 14-bus models, DAD, RL-sBOED, and Step-DAD obtain higher mean sPCE information scores than Random, Fixed, and Myopic.

Estimator saturation limits comparisons among learned policies on the IEEE 14-bus model. DAD and RL-sBOED replace online planning with policy forward passes. Step-DAD adds simulation and gradient refinement, with modest score gains on IEEE 9-bus and no consistent improvement on IEEE 14-bus under the original estimator. Before concluding that these methods are ready for deployment, future work should evaluate estimator sensitivity, model mismatch, and decision latency.

REFERENCES

- [1] P. Denholm, T. Mai, R. W. Kenyon, B. Kroposki, and M. O'Malley, "Inertia and the power grid: A guide without the spin," National Renewable Energy Laboratory, Tech. Rep. NREL/TP-6A20-73856, 2020. [Online]. Available: <https://docs.nrel.gov/docs/fy20osti/73856.pdf>
- [2] P. Wall and V. Terzija, "Simultaneous estimation of the time of disturbance and inertia in power systems," *IEEE Transactions on Power Delivery*, vol. 29, no. 4, pp. 2018–2031, 2014.
- [3] J. W. Pierre, N. Zhou, F. K. Tuffner, J. F. Hauer, D. J. Trudnowski, and W. A. Mittelstadt, "Probing signal design for power system identification," *IEEE Transactions on Power Systems*, vol. 25, no. 2, pp. 835–843, 2010.
- [4] J. Peng, J. Tan, P. Koralewicz, E. Mendiola, S. Dong, A. Hoke, and K. Horowitz, "Probing signal-based inertia and frequency response estimation for power systems with high levels of inverter-based resources," in *2024 IEEE Power & Energy Society General Meeting (PESGM)*, 2024, pp. 1–5.
- [5] H. Hjalmarsson, "System identification of complex and structured systems," *European Journal of Control*, vol. 15, no. 3–4, pp. 275–310, 2009.
- [6] X. Huan and Y. M. Marzouk, "Sequential Bayesian optimal experimental design via approximate dynamic programming," arXiv:1604.08320, 2016. [Online]. Available: <https://arxiv.org/abs/1604.08320>
- [7] A. Foster, D. R. Ivanova, I. Malik, and T. Rainforth, "Deep adaptive design: Amortizing sequential Bayesian experimental design," in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139, 2021, pp. 3384–3395.
- [8] T. Blau, E. V. Bonilla, I. Chades, and A. Dezfouli, "Optimizing sequential experimental design with deep reinforcement learning," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162, 2022, pp. 2107–2128.
- [9] M. Hedman, D. R. Ivanova, C. Guan, and T. Rainforth, "Step-DAD: Semi-amortized policy-based Bayesian experimental design," in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267, 2025, pp. 22 904–22 923. [Online]. Available: <https://proceedings.mlr.press/v267/hedman25a.html>
- [10] R. D. Zimmerman, C. E. Murillo-Sanchez, and R. J. Thomas, "MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education," *IEEE Transactions on Power Systems*, vol. 26, no. 1, pp. 12–19, 2011.
- [11] R. Kaur and D. Kumar, "Transient stability improvement of IEEE 9 bus system using power world simulator," *MATEC Web of Conferences*, vol. 57, p. 01026, 2016.